

WeatherGPT (SIH26068)

Conversational AI for Weather Forecasting, Alerts & Climate Information

Ministry of Earth Sciences (MoES) • India Meteorological Department (IMD) • Theme: Disaster Management
Release: v1.0.0 Production Prototype • Document Date: September 2026

1. Executive Summary & Problem Formulation

India generates abundant, high-resolution meteorological intelligence via the India Meteorological Department (IMD) and Numerical Weather Prediction (NWP) centers. However, there exists a critical **'last-mile translation gap'**: raw atmospheric metrics (e.g., 28°C, 82% humidity, WMO code 65, 14 km/h gusts) are difficult for an everyday farmer, commuter, or traveler to interpret into operational decisions.

WeatherGPT bridges this gap by deploying an authoritative conversational AI layer that interprets verified meteorological feeds into clear, native-language advice (Hindi, Hinglish, English) while strictly upholding **Zero-Hallucination Grounded AI**.

Grounded AI Architecture Rule: The AI model never invents, guesses, or hallucinates weather numbers. Numerical forecasts are fetched from verified Open-Meteo & NWP APIs; the AI strictly narrates and contextualizes verified facts.

2. System Architecture & End-to-End Pipeline



3. Technology Stack & Key Specifications

Subsystem	Technology / Tool	Implementation Purpose
Backend Framework	FastAPI (Python 3.12)	High-speed asynchronous ASGI REST service with automatic Swagger UI at /docs.
Data Validation	Pydantic v2	Strict typed schemas for chat requests, weather cards, and disaster alert items.
Async Network	HTTPX	Non-blocking asynchronous HTTP client for external meteorological API calls.
AI / LLM Layer	Google Gemini 1.5 Flash + NLU	Dual engine: Gemini API if key is set, plus built-in rule-based grounded template generator.
Meteorological Feeds	Open-Meteo REST & NWP	Live current, 24-hour hourly, 7-day outlook, and ECMWF / GFS / ICON multi-model comparisons.
Doppler Radar	Leaflet.js + RainViewer API	Real-time precipitation radar tile overlays on CartoDB dark basemap.
Voice Processing	Web Speech API (STT / TTS)	Zero-latency browser-native voice recognition and audio speech narration in Hindi & English.
Frontend UI	HTML5 / CSS3 / ES6 Vanilla JS	Lightweight, zero-bundle overhead console designed for smooth performance on 2G/3G.

4. Backend Modular Service Breakdown

4.1 Conversational Router & AI Engine (chat.py & ai_service.py)

- **Intent & Dialog Classification:** Distinguishes between weather, greeting, identity, and off_topic queries.
 - **Off-Topic Safety Guardrails:** Non-weather queries (cricket, films, politics, coding) are politely deflected with a live city weather snapshot to prevent hallucination.
 - **Entity & Timeframe Extraction:** Extracts 30+ Indian cities and timeframes (today, kal, weekend) using regex and NLP patterns.
 - **Persona Auto-Inference:** Automatically routes agricultural queries (khet, sinchai, fasal) to the Kisan persona even if the user has not explicitly selected it.

4.2 Weather & Numerical Weather Prediction Service (weather_service.py)

- **Live Multi-Parameter Forecast:** Fetches temperature, apparent feels-like, humidity, precipitation, wind speed, wind direction, and UV index.
- **WMO Code Decoding:** Translates 20+ WMO atmospheric codes into localized emojis and text descriptions.
- **NWP Multi-Model Comparison:** Compares ECMWF IFS-025, NOAA GFS Seamless, and DWD ICON Seamless for 3-day max temperatures and rain totals.
- **Offline Fallback Engine:** Supplies realistic synthetic meteorology if external networks fail, preventing crashes.

4.3 Early Warning & Alert Service (alert_service.py)

- **Severe Heatwave (Red Alert):** Ambient temperature ≥ 42°C triggers emergency ORS hydration and direct sunlight avoidance notices.
- **Heavy Downpour & Squall (Orange Alert):** Rain > 25mm or rain probability > 80% with storm codes issues pesticide halt and underpass hazard warnings.
- **High Wind / Gale Warning:** Wind gusts > 45 km/h trigger crop lodging and structural hazard advisories.
- **Regional CAP Registry:** Pre-configured bulletins for high-density hubs (Mumbai Orange Alert, Indore Yellow Watch, Delhi Heat Stress).

4.4 Domain Persona Advisory Engine (advisory_engine.py)

- **■ Kisan (Farmer):** Advises irrigation deferral if rain probability exceeds 50%; warns against foliar spray drift if wind exceeds 18 km/h.
- **■ Yatri (Traveler):** Evaluates asphalt hydroplaning risk; recommends low-beam fog lights under low visibility conditions.
- **■ Nagrik (Citizen):** Recommends carrying an umbrella; warns of UV hazards when index exceeds 7.0.
- **■ Disaster Officer:** Formulates culvert waterlogging warnings and triggers SDRF Level-2 stand-by readiness notices.

5. REST API Endpoints Specification

Endpoint	Method	Parameters	Response Model & Summary
/api/chat	POST	message, persona, language, city, lat, lon	ChatResponse: answer, weather_card, alert, advisories, followups, sources.
/api/weather/forecast	GET	location, lat, lon	WeatherCardData: current, 24h hourly points, 7-day daily forecast.
/api/weather/models	GET	location	NWPComparisonResponse: ECMWF vs GFS vs ICON comparison & consensus.
/api/alerts/active	GET	None	List[AlertItem]: Active regional CAP bulletins across monitored districts.
/api/climate/history	GET	location	ClimateTrendResponse: 5-year rainfall and temperature historical trend analysis.
/health	GET	None	Health status: { 'status': 'healthy', 'version': '1.0.0' }

6. Frontend Interaction & Dashboard Console

The frontend (frontend/index.html) is divided into two synchronized consoles:

- 1. Conversational Console (Left):** Features Web Speech voice mic input (STT), text-to-speech audio button (TTS), alert tags, dynamic suggestion follow-up chips, and an expandable 'Why This Answer?' (*Grounding Transparency Inspector*) documenting the met grid point, verified feeds, and DPDP Act compliance. Includes a direct client-side fallback engine to query Open-Meteo if the backend is down.
- 2. Visual Weather & Risk Console (Right):** Provides 4 switchable tabs: *Live Metrics* (hero card, 24h visual risk ribbon, dynamic persona intelligence panels for Kisan/Yatri/Officer, hourly scroll, and 7-day outlook), *Doppler Radar* (interactive Leaflet map with RainViewer live precipitation tiles), *NWP Models* (ECMWF vs GFS vs ICON cards), and *Climate Trends* (5-year rainfall variability bar chart).

7. As-Built Data Baseline & Compliance Map

As cataloged in DATA_BASELINE.md, here is the transparent status of all datasets:

Pipeline	Current Status	Data Source	Production Integration Plan
Forecast (Current, 24h, 7d)	100% Live	Open-Meteo REST API	Live global NWP; production ready for IMD Bharat Forecast System (BFS).
Location / Geocoding	100% Live	Open-Meteo + In-Memory DB	Live geocoding with instant fallback for 25+ major Indian cities.
Doppler Weather Radar	100% Live	RainViewer Radar Tiles	Live precipitation radar scan stream on dark basemap.
NWP Model Comparison	100% Live	Open-Meteo Multi-Model	Live queries for ECMWF IFS-025, NOAA GFS, and DWD ICON.
NDMA SACHET / CAP Alerts	Hybrid Prototype	Static CAP + Rules Engine	Simulated bulletins for Mumbai/Indore/Delhi + dynamic threshold evaluator.
Air Quality (AQI)	Baseline (45)	Schema Default Placeholder	Baseline ready for direct CPCB National Air Quality Index API plug-in.
5-Year Climate Trends	Climatology Baseline	Latitude Formula Baseline	Regional latitude climatology baseline for monsoon trend analysis.

8. How to Execute & Deploy

```
# 1. Start FastAPI Backend (Port 8000)
cd weathergpt/backend
.\.venv\Scripts\activate
uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
# Swagger UI: http://localhost:8000/docs

# 2. Start Frontend Static Server (Port 3000)
cd weathergpt/frontend
python -m http.server 3000
# Web Console: http://localhost:3000
```